

Mahir Ashab

Chandpur-3600, Bangladesh

ashabmahir.200@gmail.com | +8801768363022

mahirashab.me | github.com/Drogalion01 | linkedin.com/in/ashab-mahir

Professional Summary

Final-year Computer Science & Engineering student and applied AI researcher building data and AI systems for low-resource, real-world settings, including **Bangla language models, edge AI for precision agriculture, and multimodal deepfake detection for digital trust**. Three research papers submitted to **ICCIT 2026**, with backend engineering experience in **Python, FastAPI, PostgreSQL, and REST APIs** and leadership of two university clubs. Interested in applying data and AI to sustainable development and digital innovation.

Technical Skills

- **Programming & Data:** Python, SQL, NumPy, pandas, Matplotlib
- **Machine Learning & AI:** PyTorch, scikit-learn, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Multimodal Learning, Model Compression, Explainable AI
- **Models & Methods:** Random Forest, ResNet50, Wav2Vec2 XLS-R, BanglaBERT, MaxViT, SHAP, Grad-CAM
- **Backend & Platforms:** FastAPI, PostgreSQL, REST APIs, Flutter, Firebase, Git, GitHub, Linux, Docker, Raspberry Pi, Arduino

Research Experience

SmartAgro: Edge Intelligence for Crop Recommendation, Disease Detection, and Adaptive Irrigation

AI & Embedded Systems Researcher

Submitted to ICCIT 2026; awaiting review decision.

- Developed a **fully local Raspberry Pi–Arduino edge AI system** for crop recommendation, paddy disease detection, environmental sensing, and automated irrigation without dependence on continuous cloud connectivity.
- Built a **Random Forest + SHAP** crop recommendation model across 22 crop classes and integrated sensor-driven inference into the local decision pipeline.
- Fine-tuned **MaxViT Tiny** for 10-class paddy disease detection, achieving **97.60% single-pass test accuracy**, and deployed image inference locally on Raspberry Pi for resource-constrained operation.

FakeBDYouth: A Bilingual Deepfake Dataset of Bangladeshi Youth with Multimodal Detection

AI Researcher

Submitted to ICCIT 2026; awaiting review decision.

- Developed a Bangla–English audiovisual deepfake dataset with **3,345 clips from 30 participants** across four authenticity classes.
- Built a multimodal detection pipeline using **ResNet50, Wav2Vec2 XLS-R, cross-modal attention, and gated feature fusion**.
- Evaluated performance using **5-fold speaker-independent cross-validation**, achieving **82.18% fusion accuracy, 88.20% manipulation-class F1, and 83.98% AUC**.

Evaluating Performance Recovery and Domain Robustness in INT8-Quantized BanglaBERT

AI Researcher

Submitted to ICCIT 2026; awaiting review decision.

- Evaluated **BanglaBERT** using unstructured pruning, dynamic INT8 quantization, cross-entropy recovery, and knowledge distillation.
- Measured predictive performance, model size, CPU inference latency, serialization stability, and out-of-domain robustness.
- Conducted experiments using **45,884 training samples**, a held-out in-domain test set, and two out-of-domain datasets.

Software Development Experience

Resume Pilot

Backend Developer

- Developed a resume analysis and interview management system using **FastAPI** and **PostgreSQL**.
- Built **10+ RESTful API endpoints** to automate resume screening and reduce manual steps in the candidate evaluation workflow.
- Designed database schemas and integrated backend services with a **Flutter** frontend for end-to-end candidate workflows.

AutomateX

Full Stack Developer

- Developed a content management system using **Flutter** and **Firebase** with real-time synchronization, authentication, and role-based access.

Fire Fighting Robot

Embedded Systems Developer

- Built an autonomous fire detection and extinguishing robot using **Arduino** and **C++**, integrating sensing, navigation, and actuator control.

Education

B.Sc. in Computer Science & Engineering

Feb 2023 – July 2027 (Expected)

Patuakhali Science & Technology University

CGPA: 3.36 / 4.00

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems, Object-Oriented Programming, Software Engineering

HSC in Science

2018 – 2019

Notre Dame College, Dhaka

GPA: 5.00 / 5.00

Languages

Bangla (Native) | English (Fluent)

Leadership Experience

PSTU Career Club

President

Mar 2026 – Present

- Lead a team of **20+ members** and coordinate career development, placement, networking, and university-industry engagement initiatives.

PSTU Entrepreneurship Club

President

Feb 2026 – Present

- Direct entrepreneurship and innovation programs connecting **50+ students** with industry mentors and stakeholders.

PSTU Career Club

Head of Event & IT Management

May 2025 – Mar 2026

- Organized **Career Fest 2025** with 5 events, 12 sessions, and 200+ participants.

Additional Leadership: Hult Prize at PSTU – Head of Event Management | Campus Ambassador, bdapps | Asst. General Secretary, CSE Club | President, NDC English Club

Achievements

- **ICPC Regional 2024:** Ranked 363rd out of 1,800+ teams (Top 20%); also participated in **ICPC Regional 2023**.
- Awarded **Excellence in Management** for leadership and organizational contribution.